20251029EB_MN (seahorse, MitoSox, vLH78, vLH80)

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Status: In progress
Entry Created On: 18 Nov 2025 16:59:57 UTC
Entry Last Modified: 26 Nov 2025 20:47:49 UTC
Export Generated On: 01 Sep 2026 18:26:31 UTC

2025年10月29日 星期三

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年10月31日 星期五

T2 -> T3

2025年11月2日 星期日

T3 -> T4 (~14 mL for over-weekend)

2025年11月4日 星期二

Add 4mL more T4

2025年11月5日 星期三

T4 -> T5

2025年11月7日 星期五

T5 -> T6

2025年11月10日 星期一

T6 -> T7
Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)

8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info

	A	B	C
1	cell	passage	well#
2	WT11	P2	5

Table1

	A	B
1	run1	volume (mL)
2	0.025% trypsin	1.5
3	HI-FBS/CTWM	3

Table3

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	3.65	100	2	7.3	A	96	90	50	1233
3					B	24	233	12	

Plate A

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												

Plate B

	1	2	3	4	5	6
A–D	no virus	vLH80 12uL in total	vLH78 3uL in total			

2025年11月11日 星期二

Change T7 -> T7 if added virus

2025年11月13日 星期四

T7 -> T8

2025年11月15日 星期六

1/2 T8 -> 1/2 T8

2025年11月17日 星期一

1/2 T8 -> 1/2 T8
Plate B
Treat C1 and D1 with 2.5mM gluc and 0.15mM gluc, respectively, at around 18:20.

2025年11月18日 星期二

Plate A

Treat Plate A for 24h treatment at 11:15 am

seahorse_treatment

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B		2.5mM gluc 1h + 1h					0.15mM gluc 1h + 1h					
C		2.5mM gluc 1h										
D		0.15mM gluc 1h										
E		2.5mM gluc 24 h										
F		0.15mM gluc 24h										
G												
H												

Plate B

MitoSOX

1. Prepare 5uM MitoSOX red: dilute 1.2 uL 5mM mitoSOX + 1.2 uL 20uM Hoechst into 1.2 mL HBSS (warm up) at 17:30
2. treat C1 and D1 with 400uL 5uM MitoSOX red + 20nM Hoechst for 10min in HBSS and wash thrice with HBSS at 18:10 - 18:20.
3. Add 1uM Hoechst from stock since I didn't see Hoechst signal,
4. treat A1 and B1 with 2.5mM gluc and 0.15mM gluc of full Neurobasal-A at 18:15-19:15
5. Image C1 and D1 in 20 min. (C1: 19:06; D1:
6. treat A1 and B1 with 400uL 5uM MitoSOX red + 5ug/ml Hoechst in HBSS for 10 min at 19:15-19:25
7. image A1 and B1 in 30 min. (19:30 - 20:00)

Table4

	A	B	C
1	405	5%	100ms
2	560	20%	200ms
3	zstack	10um in total	1um/step

Table2

	A	B	C
1	image	treatment	treat_time
2	mitosox_MN_100x_1xzoom_001	2.5mM gluc	24h
3	mitosox_MN_100x_1xzoom_002	0.15mM gluc	24h
4	mitosox_MN_100x_1xzoom_003	2.5mM gluc	1h
5	mitosox_MN_100x_1xzoom_004	0.15mM gluc	1h

2025年11月19日 星期三

1/2 T8 -> 1/2 T8

Plate A Seahorse

450mM 2DG in 1mL media: 73.8mg in 1mL (999.03uL) media, 75.4mg in 1.021 mL media

condition: 1.5uM oligomycin, 1uM FCCP, 0.5uM Rot/AA

Bring to Pacold lab: 2.5mM and 0.15mM media, resevouirs, cells, seahorse media, laptop, notebook

Pretreat cells at 9:30-10:30

Prepare media at 9:00: seahorse media, L-glutamine, glucose, sodium pyruvate, HEPES

Bring everything to Pacold lab at 9:40.

Prepare drugs: 10:00-10:30

Change media at 10:30

Put cells on machine on 11:30

1. Rehydrate the cartridge the day before.
2. Turn on the seahorse machine.
3. Prepare drugs from stock as the drug table below with two glucose concentrations (with or without HEPES)
4. every well wash with ~70 uL media with corresponding glucose concentration
5. add 175uL of media to the cells and add 2.5mM glucose media to all the blank wells.
6. incubate the plate for 1 hour to deplete CO2.
7. Take out the cartridge and remove the pink part.
8. add drug to the cartridge as below. Add H2O for all the remaining wells.

drug in cartridge layout_mito stress

	A	B
1	A (oligo)	B (FCCP)
2	C (Rot/AA)	D (H2O)

drug in cartridge layout_GR

	A	B
1	A (Rot/AA)	B (2DG)
2	C (H2O)	D (H2O)

drug calculation per glucose concentration

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1			stock (uM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution	**working (uM)**	loading volume	well volume before loaded	loading dilution	loading conc.(uM)	dilution factor from 1st dil.	well# for each condition	total loading volume	volume of 1st dil.	volume of media	
2	mito stress	oligo	10000	10	20	2	18	1000	**1.5**	25	175	8	12	83.3333333333	15	425	5.1	419.9	
3		FCCP	10000	20	20	1	19	500	**1**	25	200	9	9	55.5555555556	15	425	7.6	417.4	
4		Rot/AA	10000	10	10	1	8	1000	**0.5**	25	225	10	5	200	15	425	2.1	420.8	
5	glycolysis rate	Rot/AA	10000	10	10	1	8	1000	**0.5**	25	175	8	4	250	10	300	1.2	297.6	
6		2DG	450000					500000	**50000**	25	200	9	450000	1	10	300	270.0	30.0	
7		medium								275				total occupied wells (Mito)	15	4675		4675	5933
8										450				total occupied wells (GR)	10	5400		5400	6727.6
9							45								8640			11660.6	
10															20345.6			11705.6	

use 2mM glutamine for this time (instead of 4mM)

prepare seahorse basal media

	A	B	C	D	E	F	G
1		stock (mM)	**working (mM)**	loading dilution	total loading volume	volume of media	
2	glucose	1100	**2.5**	440	22000	50.0	21730.0
3		1100	0.15	7333.3333333333	13000	1.8	12868.2
4	glutamine	200	**2**	100	22000	220.0	
5		200	**2**	100	13000	130.0	
6	HEPES	1000	5	200	7000	35.0	6965

Cell conditions

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B										bad cells (flushed)		
C												
D		bad cells (not attached)										
E												
F										bad cells (flushed)		
G												
H												

assay map

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B		2.5mM mito					0.15mM mito					
C		2.5mM mito					2.5mM GR					
D		0.15mM mito					0.15mM GR					
E		2.5mM mito					2.5mM GR					
F		0.15mM mito					0.15mM GR					
G												
H												

2025年11月20日 星期四

Plate B
treat 24h at 17:30

2025年11月21日 星期五

new spinning disk confocal
observation: the nucleus look a bit bigger and more round with vLH80 compared to vLH78

78_O.C. , 100X, 1xzoom, 600x900, 12-bit

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	488	12	100	10ms	
3	640	12	10	100 ms	snapshot
4	488	12	30	50ms	

80_O.C. , 100X, 1xzoom, 600x900, 16-bit

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	10	200 ms	
3	640	16	10	100 ms	

Table5

	A	B	C	D	E	F	G	H	I
1			vLH78			vLH80			
2	condition#	condition	treat time	image time	total position	treat time	image time	total position	
3	1	2.5mM glucose 1h	16:55	18:10	12+3	17:57	19:05	20	
4	2-1	0.15mM glucose 1h	17:15	18:25	16	18:13	19:20	20	
5	2-2	2.5mM glucose 24h	~17:30 11/20	17:20-17:50	15	~17:30 11/20	19:40	20	
6		0.15mM glucose 24h	~17:30 11/20	18:50	15	~17:30 11/20	19:55	20	

fileID_treatment_78

	A	B	C	D	E
1	image	treatment	treat time	type	
2	vLH78_MN_100x_1xzoom_0001	2.5mM gluc	24h	movie	
3	vLH78_MN_100x_1xzoom_0002	2.5mM gluc	24h	snapshot	
4	vLH78_MN_100x_1xzoom_0003	2.5mM gluc	24h	movie	some were out of focus. Retake them.
5	vLH78_MN_100x_1xzoom_0004	2.5mM gluc	24h	snapshot	
6	vLH78_MN_100x_1xzoom_0005	2.5mM gluc	1h	movie	
7	vLH78_MN_100x_1xzoom_0006	2.5mM gluc	1h	snapshot	
8	vLH78_MN_100x_1xzoom_0007	0.15mM gluc	1h	movie	
9	vLH78_MN_100x_1xzoom_0008	0.15mM gluc	1h	snapshot	
10	vLH78_MN_100x_1xzoom_0009	0.15mM gluc	24h	movie	
11	vLH78_MN_100x_1xzoom_0010	0.15mM gluc	24h	snapshot	

fileID_treatment_80

	A	B	C	D
1	image	treatment	treat time	
2	vLH80_MN_100x_1xzoom_0001	2.5mM gluc	1h	
3	vLH80_MN_100x_1xzoom_0002	0.15mM gluc	1h	
4	vLH80_MN_100x_1xzoom_0003	2.5mM gluc	24h	
5	vLH80_MN_100x_1xzoom_0004	0.15mM gluc	24h	